cogs 9 | Week 3

Data wrangling

Brendan Tomoschuk

Reminders

Late adds + International students

1. Reading Quiz 1, syllabus quiz open until **FRIDAY WEEK 3**

Reading Quiz 2

Due **this** Wednesday!

Reading Quiz 3

Due **next** Wednesday!

Midterm 1 — Monday October 19

- **Format:** 35 multiple-choice questions, 50 mins, ***CLOSED-NOTE***
  - ***Questions taken from***
    - ***lectures up to and including “Wrangling data”***
    - ***Readings 1-3***
- **Bring:**
  - A pen or pencil
  - No need to bring a scantron
- **Reminders:**
  - Come early and get a seat to maximize everyone’s time!
  - Students with accommodations will take the exam at the Triton Testing Center. Email me ASAP if you have not sent me your accommodations letter

Midterm 1 — Monday October 19

- **Looking for volunteers for the overflow room, CSB 180. Smaller group setting in the Cog Sci building**



Group Project Guidelines

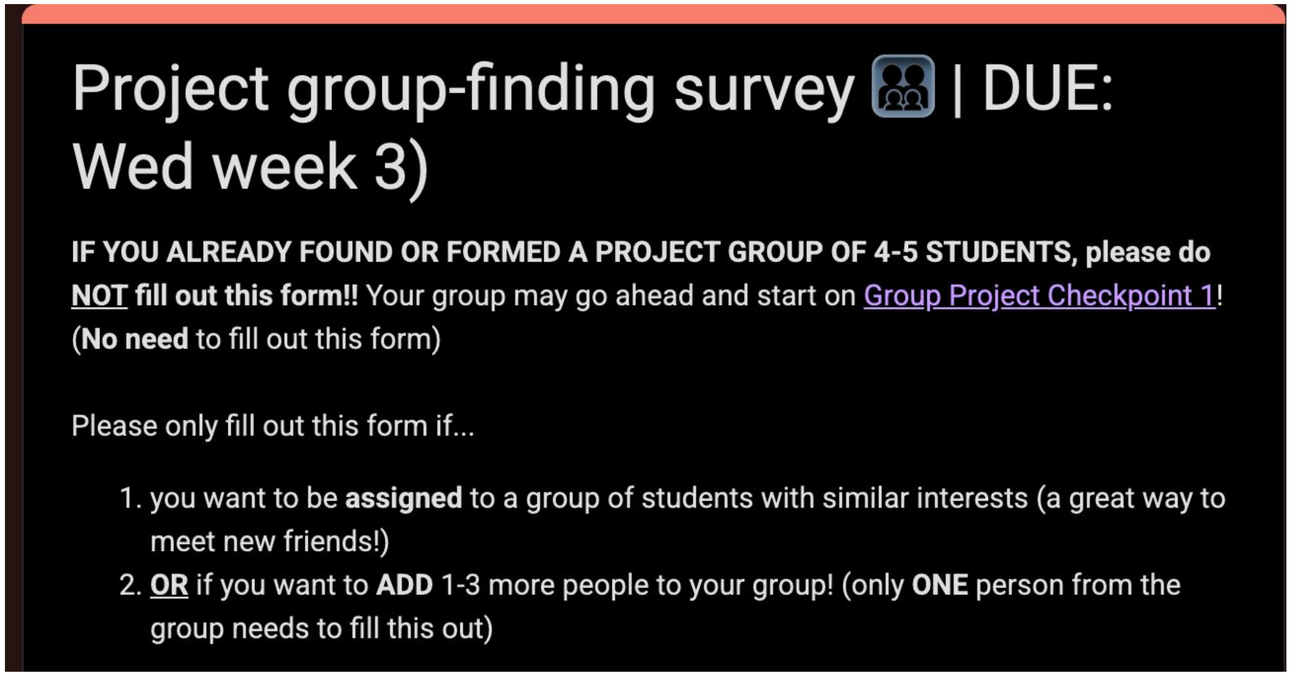
Groups of 4-5 students

1. Form your own group **OR**
2. Be matched with a group based on interests / work styles

Group Project Checkpoint 1 is

DUE Wednesday of Week 5

- If you want to be matched **OR** want 1-3 more people in your group, please fill out the group-finding survey by **Wednesday of week 3!**
  - We will match groups by the **end of that week BY EMAIL** **(contact your group ASAP)**





ImageNet: The dataset that changed the development of AI



Dr. Fei-Fei Li (李飞飞)

Stanford University

Set up by data scientist Fei-Fei Li in 2006, the ImageNet database now contains more than 14 million annotated images. It has played a key role in advancing computer vision across applications like object recognition, image classification, and object localization.

Tidy Data

1. Each variable forms a column
2. Each observation forms a row
3. Each type of observational unit forms a table



Dr. Hadley Wickham

RStudio, creator of Tidyverse



Music analysis

You are analyzing music streaming across three platforms: Spotify, Apple Music, Youtube.



How do you organize your data:

1. What is your data science question?
2. What is the observational unit (what does each row represent?)
3. What are the variables?
4. [bonus] How many tables do you have and what do they represent?





(~5-7 min with your neighbors)

Tidy spreadsheets

1. Be consistent
2. Choose good names for things
3. Write dates as YYYY-MM-DD (ISO-8601)
4. No empty cells
5. Put just one thing in a cell
6. Don't use font color or highlighting as data
7. Save the data as plain text files (e.g., CSV)

Be consistent

Keep the same variable names across different spreadsheets / tables

user_id	name	age	sex
1	Marcus J.	29	M
2	Aisha Khan	34	F
3	Lee Chen	22	F

Sex is specified here as “M” or “F”

Pick a way to code your variables and stick to it!

(e.g., don’t enter “female” for some data and “F” for others)

Choose good names for things

Guideline	Do this...	Not this!
Avoid Extra Spaces	‘male’	‘male ’
Use underscores, not spaces	doctor_visit_1	Doctor Visit 1
Choose meaningful names	doctor_visit	“F1”

Write dates as YYYY-MM-DD (ISO-8601)

- Different countries have different ways of writing the date!
  - **USA:** MM/DD/YY
  - **Most of Europe:** DD/MM/YY
  - **China/Japan:** YYYY年 MM月 DD日; **Korea:** YYYY년 MM월 DD일
- **International Standard ISO 8601** specifies numeric representations of date and time
  - Largest unit to smallest: **YYYY-MM-DD TZ** (timezone) **HH:MM:SS**
- Standard notation helps to avoid confusion in international communication! (And so your data can be used worldwide!)

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course code:
WSTAPX

What should we do if there are missing values in our data?

person	treatment	result
John Smith	a	—
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11
Mary Johnson	b	1

(A) Replace it with a missing-value identifier

(B) Replace it with a 0

(C) Do nothing

(D) A or B works

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(A) Replace it with a missing-value identifier

(B) Replace it with a 0

(C) Do nothing

(D) A or B works

No empty cells

Replace missing data cells with a **consistent, missing-value identifier** (e.g., NaN or NA, **NOT** 0!)

usage_id	user_id	date	device_os	screen_time_min	emotion_rating
1	1	2026-01-15	iOS		3
3	2	2026-01-15	Android	58	
4	3	2026-01-16	Android	15	1

usage_id	user_id	date	device_os	screen_time_min	emotion_rating
1	1	2026-01-15	iOS	**NaN**	3
3	2	2026-01-15	Android	58	**NaN**
4	3	2026-01-16	Android	15	1

Put just one thing in a cell

Don't encode the unit in the same cell as numbers!

user_id	name	age	sex	weight
1	Marcus J.	29	M	180 lbs
2	Aisha Khan	34	F	120 lbs
3	Lee Chen	22	F	150 lbs

Put the units in the variable name

user_id	name	age	sex	weight_lbs
1	Marcus J.	29	M	180
2	Aisha Khan	34	F	120
3	Lee Chen	22	F	150

Don’t use font color or highlighting as data

Don’t highlight things like outliers

user_id	name	age	sex
1	Marcus J.	92	M
2	Aisha Khan	34	F
3	Lee Chen	22	F

Make a new indicator variable for outliers

user_id	name	age	sex	outlier
1	Marcus J.	92	M	TRUE
2	Aisha Khan	34	F	FALSE
3	Lee Chen	22	F	FALSE

Save the data as plain text files (e.g., CSV)

- **Portable & long-lasting!** Plain text can be **opened on any computer** and will still work years later
- **Nearly every tool** (Excel, R, Python, SQL systems) can read it!
- Good for **version control:** Git can track line-by-line changes in text files; not so well in other binary formats like .xlsx.
- **Avoids hidden spreadsheet behavior** like auto-formatting values (e.g., converting IDs to dates)

Data dictionaries

Data dictionary: a separate file (usually a reference table) that explains the data and variables

- Exact variable names (matching the columns)
- Description (what the variable means)
- Units (e.g., hours, dollars, bpm)
- Expected range / valid values

Example: Data dictionary

variable_name	display_name	description	type	units	allowed / expected values
user_id	User ID	Unique identifier for each participant	integer	none	positive integers; unique
name	Name	Participant name (or pseudonym)	string	none	text
age	Age	Participant age at start of study	integer	years	0–120 (expected 18–100)
sex	Sex	Self-reported sex	categorical	none	{F, M, Nonbinary, Prefer_not_to_say}

Messy Data

1. Column headers are values but should be variable names
2. A single column has multiple variables
3. Variables have been entered in both rows and columns
4. Multiple "types" of data are in the same spreadsheet
5. A single observation is stored across multiple spreadsheets

Messy Data

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Column headers are values but should be variable names

religion	<$10k	$10–20k	$20–30k	$30–40k	$40–50k	$50–75k
Agnostic	27	34	60	81	76	137
Atheist	12	27	37	52	35	70
Buddhist	27	21	30	34	33	58
Catholic	418	617	732	670	638	1116
Don't know/refused	15	14	15	11		
Evangelical Prot	575	869	1064	982		
Hindu	1	9	7	9		
Historically Black Prot	228	244	236	238		
Jehovah's Witness	20	27	24	24		
Jewish	19	19	25	25		

religion	income	freq
Agnostic	<$10k	27
Agnostic	$10–20k	34
Agnostic	$20–30k	60
Agnostic	$30–40k	81
Agnostic	$40–50k	76
Agnostic	$50–75k	137
Agnostic	$75–100k	122
Agnostic	$100–150k	109
Agnostic	>150k	84
Agnostic	Don't know/refused	96

A single column has multiple variables

country	year	column	cases
AD	2000	m014	0
AD	2000	m1524	0
AD	2000	m2534	1
AD	2000	m3544	0
AD	2000	m4554	0
AD	2000	m5564	0
AD	2000	m65	0
AE	2000	m014	2
AE	2000	m1524	4
AE	2000	m2534	4
AE	2000	m3544	6
AE	2000	m4554	5
AE	2000	m5564	12
AE	2000	m65	10
AE	2000	f014	3

(a) Molten data

country	year	sex	age	cases
AD	2000	m	0–14	0
AD	2000	m	15–24	0
AD	2000	m	25–34	1
AD	2000	m	35–44	0
AD	2000	m	45–54	0
AD	2000	m	55–64	0
AD	2000	m	65+	0
AE	2000	m	0–14	2
AE	2000	m	15–24	4
AE	2000	m	25–34	4
AE	2000	m	35–44	6
AE	2000	m	45–54	5
AE	2000	m	55–64	12
AE	2000	m	65+	10
AE	2000	f	0-14	3

(b) Tidy data

Variables have been entered in both rows and columns

id	year	month	element	d1	d2	d3	d4	d5	d6	d7	d8
MX17004	2010	1	tmax	—	—	—	—	—	—	—	—
MX17004	2010	1	tmin	—	—	—	—	—	—	—	—
MX17004	2010	2	tmax	—	27.3	24.1	—	—	—	—	—
MX17004	2010	2	tmin	—	14.4	14.4	—				
MX17004	2010	3	tmax	—	—	—	—				
MX17004	2010	3	tmin	—	—	—	—				
MX17004	2010	4	tmax	—	—	—	—				
MX17004	2010	4	tmin	—	—	—	—				
MX17004	2010	5	tmax	—	—	—	—				
MX17004	2010	5	tmin	—	—	—	—				

id	date	tmax	tmin
MX17004	2010-01-30	27.8	14.5
MX17004	2010-02-02	27.3	14.4
MX17004	2010-02-03	24.1	14.4
MX17004	2010-02-11	29.7	13.4
MX17004	2010-02-23	29.9	10.7
MX17004	2010-03-05	32.1	14.2
MX17004	2010-03-10	34.5	16.8
MX17004	2010-03-16	31.1	17.6
MX17004	2010-04-27	36.3	16.7
MX17004	2010-05-27	33.2	18.2

(b) Tidy data

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course code:

WSTAPX

Which one of these is NOT a messy spreadsheet?

(A)

name	test1	test2	test3
Alice	88	91	85
Bob	92	89	90

(B)

student	exam	score
Alice	1	88
Alice	2	91
Bob	1	92
Bob	2	89

(C)

student	exam1_score	exam2_score
Alice	88	91
Bob	92	89

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Which one of these is NOT a messy spreadsheet?

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Alice	88	91	85
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(B)

student	exam	score
Alice	1	88
Alice	2	91
Bob	1	92
Bob	2	89

(C)

student	exam1_score	exam2_score
Alice	88	91
Bob	92	89

What's wrong with how this dataset is organized?
How would you fix it using tidy data/spreadsheet principles?

(~5 min with your neighbor)

	A	B	C	D	E
1		marcus j. Pettigew (m)	penny (f)	Jerry Smith	Jessica Chun (f)
2	3/22/2019	2 hours	3		2
3	3/23/2019	3.5 hours	4.6		1.11
4	3/24/2019	3.1 hrs	1	2.2	4.2
5	3/25/2019	**33 hours**	1.5	2	2
6	3/26/2019	0.5 hours	0	5.3	

...

F	G	H	I	J	K	L	M
marcus	penny	jerry	jessica	marcus	penny	jerry	jessica
ios	android	android	ios	good	fine	ok	not bad
ios	android	android	ios	ok	fine	ok	good
ios	android	android	ios	great	fine		good
ios	android	android	ios	great	fine	sad	good
ios	android	android	ios	depressed	fine	happy	meh

What's wrong with this dataset?

1. Columns are not treated as variables (name, sex, phone OS, date, screen time, emotion rating)
2. Rows are are not treated as observations
3. Datetime in wrong format
4. Sex encoded in column header
5. Repeated column headers
6. 1 table, but 4 types of data! (user data, screen time data, emotional data, datetime data)
7. Outliers underlined
8. Missing data highlighted
9. Vague emotional ranking (potential issue)
10. No title

Our wrangling job

1. Columns are now variables (user_id, name, sex, phone_os, date, screen_time_hours, emotion)
2. Rows are observations (date and user)
3. Datetime formatting fixed
4. Sex no longer encoded in column header
5. Columns are only repeated through tables
6. 2 tables: one for user info, one for the study (screen time with emotion)
7. Outliers encoded with NaN
8. Missing data encoded with None
9. We recoded emotion on a qualitative scale (1–5)
10. Our file names are our tables' titles

user_information.csv

user_id	name	sex	phone_os
0	marcus j. Pettigew	m	ios
1	penny	f	android
2	Jerry Smith	None	android
3	Jessica Chun	f	ios

user_screen_time_emotion.csv

date	user_id	screen_time_hours	emotion
2019-03-22	0	2	1
2019-03-22	1	3	3
2019-03-22	2	Nan	2
2019-03-22	3	2	5
2019-03-23	0	3.5	5
2019-03-23	1	4.6	3
2019-03-23	2	Nan	None
2019-03-23	3	1.11	3
2019-03-24	0	3.1	1
2019-03-24	1	1	1
2019-03-24	2	2.2	1
2019-03-24	3	4.2	2
2019-03-25	0	NaN	3
2019-03-25	1	1.5	2
2019-03-25	2	2	1
2019-03-25	3	2	3
2019-03-26	0	0.5	4
2019-03-26	1	0	5
2019-03-26	2	5.3	5
2019-03-26	3	NaN	2

Takeaways

- Follow tidy data / spreadsheets principles!
- Tidy principles…
  - make your data easy for many people to use
  - make later steps in the data science process (e.g., analysis, modeling) faster, more reliable, and more reproducible!